

NEXUS-1 Glossary

A plain-English guide to every term in the console - no nuclear background needed

How to use this glossary

NEXUS-1 is a demonstration console for a nuclear power plant. It is a simulation - nothing is connected to a real reactor, and the models are studied and representative, not certified engineering tools. This glossary explains the words you will meet in the menus and on the screens, written for someone with no nuclear training. Terms are grouped by theme so related ideas sit together. Where a colour code appears in the app: green means nominal (normal), amber means degrading, red means a limit reached or the reactor tripped.

The platform and its operating modes

Digital twin. A live virtual copy of a physical system that is continuously fed with the system's own data, so the copy behaves like the real thing. NEXUS-1 is built around this idea.

Telemetry. Measurements streamed from sensors (temperatures, flows, pressures). In the app, 'telemetry' views are the simulated equivalent of real sensor feeds.

Simulation Mode. The whole plant runs virtually - useful for study and demonstration. This is what NEXUS-1 actually is.

Training Mode. A mode meant for teaching operators and rehearsing incidents safely.

Shadow Mode. The software mirrors a live plant's data but cannot act on it - it only watches.

Advisory Mode. The software offers recommendations; humans make every decision.

Supervisory Mode. The software is connected to control infrastructure for monitoring and coordination, still under human supervision.

Hybrid Digital Twin Mode. Keeps the simulation continuously in step with live plant data.

No control authority. A core safety principle here: the software may watch and advise, but it can never directly operate the reactor's safety systems. Those stay separate, certified and human-supervised.

Reactor physics and neutronics

Reactor core. The central region where nuclear fission produces heat.

Fission. The splitting of heavy atomic nuclei (like uranium), which releases heat and more neutrons - the basic energy source.

Fuel assembly. A bundle of fuel rods. The core is built from many assemblies arranged in a grid; the Core Map shows them.

Burnup. A measure of how much of the fuel has been 'used up' over time.

Neutron flux. How many neutrons are flying through the core - essentially a measure of how intensely the reaction is running. Roughly tracks reactor power.

Reactivity. How far the reactor is from steady. Positive reactivity means power tends to rise, negative means it falls, zero means it holds steady.

Critical / subcritical / supercritical. Critical = the chain reaction is exactly self-sustaining (steady power). Subcritical = it is dying down. Supercritical = it is increasing. 'Critical' here is normal and good, not dangerous.

pcm (per cent mille). A very small unit of reactivity - one hundred-thousandth. Operators nudge reactivity by a few pcm at a time.

Beta and the 'dollar'. A small fraction of neutrons are 'delayed' (see below); that fraction is called beta. One 'dollar' of reactivity equals beta. Reaching one dollar ('prompt critical') is hazardous, because the reactor would race ahead on instant neutrons alone.

Delayed neutrons / precursors. A few percent of neutrons are released seconds to minutes after fission, by fragments called precursors. This delay is what makes a reactor slow enough to control.

Prompt neutrons / generation time. Most neutrons appear almost instantly; the average time from one neutron generation to the next is tiny (around twenty millionths of a second).

Point kinetics. The standard simplified set of equations describing how reactor power changes over time. The Reactor Kinetics screen solves the real version of these.

Reactor period. The time it takes for power to change by a factor of about 2.72. A long period means a slow, gentle change; a short period means a fast one.

Doppler feedback. A built-in safety effect: as the fuel heats up it absorbs more neutrons, which automatically pushes power back down. Fast and stabilising.

Moderator temperature coefficient. A similar self-correcting effect coming from the temperature of the coolant/moderator (the water).

Xenon-135. A strong neutron-absorbing 'poison' created as a by-product of fission. After a power reduction it keeps building for hours, which can make restarting difficult - the 'xenon pit'.

Iodine-135. A fission product that decays into xenon-135; it drives the slow xenon build-up after a shutdown.

Control and shutdown

Control rod. A neutron-absorbing rod. Pushing it into the core lowers power; pulling it out raises power.

Rod bank. A group of control rods that move together.

Insertion / withdrawal. Inserting rods = less power; withdrawing = more power. The Core Control and 3D screens let you do this.

SCRAM. An emergency shutdown: all rods drop fully into the core in seconds to stop the reaction at once. In NEXUS-1 a SCRAM also writes to the incident log and compliance record and flags the rods for inspection.

Setpoint. The target value a control is trying to reach or hold.

Heat, coolant and steam

Primary loop / coolant. Water pumped through the core to carry heat away. In a pressurised reactor it stays liquid under high pressure.

Hot leg / cold leg. The pipe carrying hot coolant out of the core (hot leg) and the cooled coolant returning (cold leg).

Loop flow. How fast the coolant is circulating.

Pressurizer. A vessel that keeps the primary coolant under high pressure so it does not boil.

Steam generator. A large heat exchanger where heat from the primary coolant boils a separate water circuit into steam. 'SG level' is the water level inside it, kept within a safe band.

Feedwater. Water pumped into the steam generator to replace what boils off. Controlled by feedwater pumps and valves (the example valve in the trip story is FV-104).

Turbine. A machine spun by high-pressure steam; it drives the generator that makes electricity.

Condenser / condenser vacuum. Cools the used steam back into water; maintaining a vacuum makes the turbine more efficient.

Governor valve. Throttles how much steam reaches the turbine.

Core / fuel temperature. How hot the coolant and the fuel are. The 3D view colours the core by temperature.

Delta-T. The temperature difference across the core. A large, unexpected delta-T can trigger an automatic shutdown.

Thermal power (MWt) vs electrical power (MWe). Heat produced (megawatts thermal) versus electricity delivered (megawatts electric). Only about a third of the heat becomes electricity; that fraction is the efficiency.

Electricity and the grid

Grid synchronisation. Matching the generator's speed and timing to the electrical grid before connecting to it.

Frequency (Hz). How many cycles per second the grid runs at (50 Hz across Europe). It must stay very stable.

RPM. Revolutions per minute - the rotation speed of the turbine and generator.

MVAR, voltage, power factor. Measures of electrical 'reactive' power, electrical pressure, and how well current and voltage stay in step.

Load following. Varying the plant's output to match changing demand on the grid, as opposed to 'baseload' (running flat out, steady).

Safety and protection

Reactor Protection System (RPS). A separate, certified system that automatically trips (shuts down) the reactor if limits are exceeded. It is independent of software like NEXUS-1.

Trip. An automatic shutdown triggered by a protection signal.

Decay heat. Heat that the fuel keeps producing even after shutdown, from radioactive decay. This is why cooling must continue after a reactor stops - and why power cannot drop instantly to zero.

Safety margin. How far a parameter is from its safety limit.

Defence in depth. Using several independent layers of protection, so no single failure is dangerous. The Zone Access diagram visualises this idea.

Containment. The sealed, robust building around the reactor that holds radioactivity in.

Vital area. The highest-security zones, such as the control room.

Fail-safe. Designed so that if something fails, the system moves to a safe state.

Radiation and health physics

Dose / dose rate. The amount of radiation received, and how fast it is received. Measured in sieverts (Sv) and thousandths, millisieverts (mSv).

Activity (becquerel, TBq). How much radioactive decay is happening per second. A terabecquerel (TBq) is a trillion such events per second.

Half-life. The time for half of a radioactive material to decay away. Longer half-life = stays radioactive longer.

Cs-137 / Co-60. Caesium-137 and cobalt-60: long-lived isotopes that dominate a plant's radioactivity, especially during decommissioning.

Radiologically Controlled Area (RCA). A zone where radiation is monitored and access is controlled.

Cumulative dose. The total radiation built up over time. It applies to people - and, in NEXUS-1, to robots, whose electronics also have a radiation limit.

Inspection and non-destructive testing (NDT)

Non-destructive testing (NDT). Checking a component for hidden flaws without cutting it open or damaging it.

Radiography. Using X-rays or gamma rays to photograph the inside of a part - like a medical X-ray for metal. The Rod Inspection screen shows a simulated radiograph.

Ultrasonic testing (UT). Using high-frequency sound waves to detect internal flaws by their echoes.

Eddy current testing (ET). Using electromagnetic fields to find cracks at or near the surface of conductive metals.

Magnetic particle testing (MT). Reveals surface cracks - but only on magnetic (ferromagnetic) materials. It does not work on the non-magnetic alloys used for fuel cladding, which is why the app shows it as 'not applicable' there.

Dye penetrant testing (PT). A coloured liquid is drawn into surface cracks to make them visible.

Visual testing (VT). Direct or remote visual examination - the simplest method.

Cladding. The thin metal tube that seals the nuclear fuel. Often a zirconium alloy, which is not magnetic.

Indication / defect. An 'indication' is a possible flaw flagged by a test; a 'defect' is a confirmed one.

Plant aging and degradation

Embrittlement. Metal - especially the reactor pressure vessel - gradually becoming more brittle from years of neutron bombardment.

Neutron fluence. The total dose of neutrons a material has accumulated over its life.

RTNDT shift. The temperature at which the vessel steel turns brittle slowly creeps upward as fluence accumulates; engineers track this shift.

Thermal fatigue. Damage from repeated heating and cooling cycles - each SCRAM, for example, adds a thermal cycle.

Usage factor. The fraction of a component's fatigue life that has been used up.

Stress corrosion cracking (SCC / IASCC). Cracking caused by the combination of mechanical stress and a corrosive environment; the irradiation-assisted form (IASCC) is accelerated by radiation.

Prestress loss. The gradual loosening of the steel tensioning that keeps containment concrete in compression.

Design life / licence extension. The lifespan a plant was designed for, and approval to operate beyond it. France's fleet-wide upgrade programme for this is called 'Grand Carenage'.

Surveillance capsule. Small metal samples kept inside the reactor and removed periodically to measure how the real materials are aging.

Decommissioning and waste

Decommissioning. Safely retiring and dismantling a plant at the end of its life.

Defueling. Removing the nuclear fuel from the reactor.

Decontamination. Removing radioactive contamination from surfaces and equipment.

Safe enclosure. A deliberate waiting period that lets short-lived radioactivity decay before heavy dismantling begins.

Dismantling. Physically taking apart the structures and components.

Site release. The end goal: the site is cleaned up enough to be released for other uses.

VLLW / LLW / ILW. Waste categories: very-low-level, low-level and intermediate-level radioactive waste. Most decommissioning waste is the least hazardous kind; the small intermediate fraction drives cost and storage.

Radiological inventory. The total amount of radioactivity present, which falls over time as material decays.

People, robots and access

Senior Reactor Operator (SRO). A licensed senior operator authorised for the broadest set of actions.

RP technician. Radiation Protection technician - manages radiological safety.

Minimum shift complement. The minimum number of qualified staff that must legally be on shift. The Staffing Stress Test checks whether absences would breach it.

Critical role. A position that must always be covered for the plant to operate safely.

Manipulator. A robotic arm used to handle objects remotely in hazardous areas.

AGV (automated guided vehicle). A self-driving vehicle used for moving loads around the site.

Mission readiness. Whether a suitable robot exists for a task that is also available, charged, and still within its radiation-dose budget.

RFID / GPS tag. Electronic tags that identify or locate people, vehicles and robots.

Zone classification / access authorisation. Which areas each person or machine is permitted to enter. The Permissions Matrix lists these; unauthorised presence is flagged as a violation.

AI, machine learning and the digital twin

AI diagnostics / predictive analytics. Forecasting failures and degradation from historical data using machine learning. These are advisory - they inform people, they do not act.

Supervised learning. A model trained on labelled past examples (e.g. cases that did or did not fail) to predict new ones.

Survival model / gradient boosting. Techniques used to estimate the probability that a component will fail within a given time.

Reinforcement learning (RL). A different kind of AI where an 'agent' learns by trial and error, improving by maximising a reward. In NEXUS-1 it trains only inside the simulation and only gives advice.

Agent / policy / reward / episode. The agent is the learner; its policy is the strategy it follows; the reward is the score it tries to maximise; an episode is one practice run.

Q-learning / tabular. A basic, transparent RL method that stores what it has learned in a readable table - so its decisions can be inspected rather than hidden.

Exploration vs exploitation. An RL agent balances trying new actions (exploration) against using the best it already knows (exploitation).

Explainability / black box. Whether you can see why a model made a decision. A 'black box' cannot be inspected; NEXUS-1 deliberately uses a readable policy instead.

Divergence. The gap between what the sensors measure (telemetry) and what the model expects. A growing gap is the digital twin's early-warning signal.

System dependencies

Dependency / influence weight. How strongly one part of the plant affects another, shown as a number from 0 (no effect) to 1 (direct effect).

Propagation. How a change or fault travels from one component through the ones that depend on it, often with a time delay.

Causal chain. The step-by-step cause-and-effect sequence behind an event - for example, the chain that led to a reactor trip.

Feedback loop. An effect that loops back to influence its own cause (such as temperature affecting reactivity, which affects temperature).

Cybersecurity (operational technology)

OT (operational technology). The computers and controllers that run physical equipment, as opposed to ordinary office IT. Protecting it is critical infrastructure security.

Zero-trust. A security approach that never trusts anything automatically and verifies every access request.

Network segmentation. Dividing networks into isolated sections so a breach in one cannot spread.

Air-gap. Physically isolating a network so it has no connection to the outside internet.

Hardware security module (HSM). A tamper-resistant device that safeguards cryptographic keys.

Multi-factor authentication (MFA). Requiring more than one proof of identity to log in.

Intrusion detection (IDS). A system that watches for and flags malicious activity.

Immutable logging / audit trail / hash seal. Tamper-evident records of who did what and when. A 'hash seal' is a digital fingerprint that proves a record has not been altered.

Units and common terms

Nominal. The normal, expected value or condition.

Transient. A temporary disturbance or change away from steady operation.

MWt / MWe. Megawatts thermal (heat) and megawatts electric (electricity).

degrees C. Degrees Celsius, the temperature scale used throughout.

PWR. Pressurised Water Reactor - the most common large reactor type; Units 1 and 2 in the fleet.

SMR. Small Modular Reactor - a smaller, factory-built reactor; Modules A, B and C in the fleet, shown with a smaller core.

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